wildtag.ai

Automated species identification for camera trap images - powered by AI, designed for ecologists.

USER MANUAL - VERSION 1.1 - JULY 2026

1. Overview

wildtag.ai processes folders of camera trap images automatically - detecting animals, classifying species, and organising results for review. It runs on your computer. wildtag ships without a model built in: you download the one you want from the Models tab the first time you use it, while connected to the internet. After a model is downloaded it is stored locally and works fully offline. DeepFaune is the recommended model for UK and European wildlife.

The workflow has four stages:

1. **Run**

   Point wildtag at an image folder, choose a model, and let it detect and classify everything.

2. **Validate**

   Browse AI predictions as a gallery, correct any mistakes, and mark batches as complete.

3. **Distribute**

   Package species folders as zips for volunteer validators, and collect their corrections when returned.

4. **Summary**

   Review detection statistics and a species breakdown for your survey.

2. Getting Started

1. **Download the installer**

   Download `wildtag_installer.zip` from the link provided (about 640 MB, since it bundles the Python runtime; models are downloaded separately later). Right-click it and select "Extract All". Inside is `wildtag_Setup.exe`.

2. **Install and launch**

   Double-click `wildtag_Setup.exe` and follow the wizard (no administrator rights needed). It creates a desktop and Start Menu shortcut. Launch wildtag from the shortcut; it opens directly with no console window. If Windows shows a security warning, click "More info" then "Run anyway" - this is normal for software that is not yet code-signed.

3. **Download a model**

   Open the Models tab. While connected to the internet, click Download next to the model you want (DeepFaune is recommended for UK and European wildlife). This is a one-time download; the model is then stored on your computer and used offline afterwards.

4. **Organise your images**

   wildtag works on a folder of JPEGs. Subfolders are supported - all images are found automatically. Each run should correspond to one survey site or session.

> wildtag never modifies your original images. All outputs are written to new files in a `validation\` subfolder.

3. Models

Models

wildtag does not come with a species model built in. The Models tab is where you download the model you want, once, while connected to the internet. After a model is downloaded it is stored on your computer and used offline from then on. Downloading a model is the first thing to do after installing wildtag.

What the Models tab shows

Each model is listed as a card giving its region, size, licence, and a short description, along with its current status:

Status	Meaning
Not installed	Available to download. Click **Download** to fetch it.
Installed	Already on this computer and ready to use offline. A **Remove** button frees the disk space if you no longer need it.
Planned	In development, not yet available. Shown for information only; it cannot be downloaded and does not appear in the Run tab.

Downloading a model

Click **Download** on the model you want. wildtag confirms the size, then downloads it with progress shown on the card. DeepFaune is about 1.1 GB and SpeciesNet about 500 MB, so allow a few minutes on a normal connection. When it finishes, the status changes to Installed.

> You only need an internet connection for this one-time download. Once a model shows as Installed it runs completely offline, so wildtag can be used in the field with no connection.

> If the Run tab's model dropdown is empty, it means no model has been downloaded yet. Open the Models tab and download one first.

Which model to choose

DeepFaune v1.4 is fast on any computer, works offline, and is well-calibrated for UK and European species - a good default for European surveys. **SpeciesNet** covers 2000+ species worldwide; choose it for global coverage or taxonomy-level results. The full global model can be slow per image, especially without a GPU, so setting a Geographic filter (your country) is recommended to speed it up. See Appendix A for details of each model.

4. Running the Pipeline

Run wildtag

Select your image folder

Click Browse and navigate to your camera trap image folder. wildtag counts all JPEG images found inside, including subfolders.

Choose a model

Select a species model from the dropdown. Each model brings its own animal detector, so there is no separate detector to choose. Only models you have downloaded on the Models tab appear here; if the dropdown is empty, download a model first.

Model	Best for
DeepFaune v1.4	UK and European surveys (recommended); fast on any computer
SpeciesNet Global	Worldwide coverage or taxonomy-level rollup; use a geographic filter to keep it fast (a GPU helps for large projects)

> Models are not bundled with wildtag. You download the one you want from the Models tab the first time you use it (DeepFaune about 1.1 GB, SpeciesNet about 500 MB), after which it works offline. Models marked "Planned" on the Models tab are in development and cannot yet be downloaded.

> **Keeping SpeciesNet fast.** SpeciesNet classifies 2000+ species worldwide, so running the full global model can take a long time per image, especially without a GPU. Set a **Geographic filter** (your country) in the Run options: it narrows SpeciesNet to species from your region and substantially speeds up processing. wildtag

> reminds you of this if you start a SpeciesNet run with no filter set. A GPU further helps for very large projects.

Set confidence thresholds

- **Detection threshold** - images where no detection exceeds this value are recorded as `empty`. Default: 0.1.
- **Species confidence** - detections below this threshold are labelled `low_confidence`. Default: 0.6. Lower this to see more predictions; raise it for stricter results.

Geographic filter

Select a region to filter out implausible species predictions. For UK surveys, select GBR. This removes predictions like wolverine or raccoon and replaces them with the next most plausible UK species. Select None to use raw model predictions.

Performance options

- **Processing device** - CPU is always available. NVIDIA GPU (CUDA) appears automatically if detected and dramatically speeds up classification.
- **CPU threads** - defaults to one less than your total CPU count, leaving one core free for the rest of your computer.
- **Checkpoint frequency** - how often wildtag saves its progress during a run. If a long run is interrupted, it resumes from the last checkpoint rather than starting over. The default suits most projects; there is rarely a need to change it.

Output options

- **Generate IDs** - adds unique `image_id` and `detection_id` columns to the CSV.
- **Prepare validation folder** - creates downscaled annotated images organised by species, ready for the Validate tab.
- **Validation image quality** - sets how large the downscaled validation images are. Higher quality makes review easier but

produces larger files and bigger volunteer packages; lower quality is faster to make and send. The default is a good balance for most surveys.

Running

Click Run wildtag. The log panel shows live progress. With DeepFaune, a few dozen images take a couple of minutes on CPU. SpeciesNet does more work per image (a full detector, classifier, and geolocation pipeline over 2000+ species), so it is slower; setting a Geographic filter (your country) narrows the candidates and speeds it up considerably, and a GPU helps further for large projects.

> If a run is interrupted (for example the computer restarts partway through sorting images into species folders), wildtag can recover. When you reopen the project it detects the incomplete validation folder and offers to finish it, keeping the images already sorted and only completing what is missing.

5. Validating Results

Validate

Validation is where you review AI predictions and correct any errors. wildtag presents images as a gallery, one species folder at a time.

Opening a validation folder

Click Browse and select your project folder or the `validation\` folder inside it - wildtag finds it automatically. The species dropdown shows only folders with images still needing review.

Reviewing images

Images appear in a grid with the AI confidence score below each one and a bounding box drawn around the detected animal. Click any image to open the correction dialog.

- If the label is correct, click **Correct**
- If wrong, select the right species from the dropdown and click **Apply correction**

To correct several images at once, hold **Ctrl** and click each one; selected tiles are highlighted in blue. Then click any selected tile to apply a single correction to them all together. This is much faster when a whole batch shares the same error.

The correction dropdown is ordered: *unidentifiable* first, then species alphabetically, then broad groups such as mustelid or lagomorph, then special labels.

If the species you need is not listed, use the **"Not listed? Add a label"** box in the correction dialog to add it. New labels are saved to `validation/custom_species.txt` in the project, stay available for the rest of the project, and are included in volunteer packages so everyone sees the same options.

Completing a batch

Once you have reviewed all images in a folder, click **Mark batch complete**. This writes your corrections to the validation CSV. The folder will not appear again in the dropdown once fully validated.

> Images containing humans are automatically pixelated in the validation folder. Original images are never modified.

Repairing a validation folder

If any tiles show "Image not found", the folder's image list has got out of step with the images actually present (this can happen with folders made by older versions of wildtag). wildtag repairs this automatically whenever a validation folder is opened, and a **Repair folders** button on the Validate tab does the same on demand. The repair is safe: it never deletes images or loses validation work, and it keeps a backup of the original list.

How validation updates the master CSV

As you validate, wildtag writes corrections to `validation.csv` files inside each species folder. When you distribute or export, these corrections are merged into the master `results_with_ids.csv`, populating the `correct_label` and `human_verified` columns. Your master CSV always reflects your latest validation work.

6. Distributing and Collecting

Distribute

The Distribute tab handles packaging validation folders to send to volunteer validators, collecting their corrections when they are returned, and exporting your favourite (hearted) images by species.

Part A - Prepare packages for validators

Select your validation folder and click **Prepare zip packages**. wildtag creates one zip file per species folder containing the annotated images, the validation spreadsheet, the wildtag app, and instructions. Each zip goes into a `distribute\` folder next to your validation folder.

Send the relevant zip to each volunteer. You might assign one species to each person, or send multiple zips to the same person for different species.

> wildtag never sends the same image twice. Each time you prepare packages, only images that still need validation and have not already been sent are included, so you can run Prepare again after adding new data without re-sending old images. Use the **Max images per zip** setting to control batch size: smaller zips are easier to email, larger ones suit a single validator working from a drive.

What the volunteer does

1. Unzip the file and double-click `Run wildtag.bat`
2. Click Browse in the Validate tab and select the unzipped folder
3. Review images, correct labels, click Mark batch complete
4.

Return the results: either zip the whole folder and send it, or send back just the `..._validation.csv` from the species subfolder (the filename does not matter)

Part B - Collect returned validated zips

When validators return their work, drop the files into the `collect\` folder inside your project folder. A returned file can be a zip of the whole folder, or just a validator's `..._validation.csv` under any name; wildtag merges by each detection's unique ID, so the filename and folder structure do not matter, and a file may contain any subset of rows. Then click **Merge all returned validations**. wildtag will:

- Show a preview: how many validated detections and corrections are coming in
- Note any rows that do not match this project (for example a file returned to the wrong project), which are ignored
- Ask you to confirm before merging
- Update your master validation CSVs and `results_with_ids.csv`
- Move processed files to `collect\processed\`; unreadable files are left in `collect\` to retry

> The first time you click Merge, wildtag creates the `collect\` folder if it does not exist and tells you where to drop the zips.

Part C - Export as Camtrap DP

Click **Export Camtrap DP** to write your validated results as a Camera Trap Data Package (Camtrap DP), the community standard for camera-trap data. wildtag creates a `camtrapdp\` folder containing `deployments.csv`, `media.csv`, `observations.csv` and a `datapackage.json` descriptor. Because it follows the standard, this package can be shared, archived, or uploaded to GBIF and other standards-based tools without any further conversion.

> Camtrap DP export (and the detection map) use a `deployment.csv` file in your project folder, with one row per camera site giving its

> name, latitude, longitude, and deployment start and retrieval dates. Use the **Save deployment.csv template** button to create a ready-to-fill template, then complete it before exporting. Without it, export still works but the deployment details and map locations will be missing.

Part D - Export favourite images

While validating you can mark standout images as favourites (the heart on each tile). The Distribute tab can then export these into a `favourites\` folder, organised into one subfolder per species, containing only the images you hearted. This is handy for pulling out the best shots of each species for reports or presentations.

7. Summary

Summary

The Summary tab gives an at-a-glance overview of a project: what was processed, how validation is progressing, where files were written, and a breakdown of detections by species. It fills in automatically when you open it.

What was processed

Run statistics for the project: how many images and detections were processed, how many frames were empty, and which model was used.

Validation progress

How many detections have been validated so far and how many remain, so you can see how much review is left across the whole project.

How many detections per species

A table of detection counts per species. After validation this reflects your corrected labels, giving a quick species summary for the survey.

Confusion matrix

Once you have validated some images, this compares the AI's original predictions against your corrections, showing where the model agreed with you and where it did not. It is a quick way to see which species the model handles well and which it tends to confuse.

Where your files have been saved

The locations of the key outputs for the project (the results CSV, the validation folder, and any exports), so you can find them without hunting through folders.

Viewing detections on a map

If your images carry GPS metadata, or you have supplied deployment coordinates, click **Launch map** to open an interactive map of your detection locations in your web browser. A dropdown on the map lets you colour sites by species.

8. Understanding the CSV Output

wildtag produces a CSV with one row per detection - not one row per image. An image with two animals produces two rows; an empty image produces one row with `NA` for species fields.

Core columns

Column	Description
`image_id`	Unique identifier for the source image. Consistent across runs on the same image.
`detection_id`	Unique identifier for this specific detection.
`relative_path`	Path to the image relative to the survey folder root.
`label`	Final species label: `empty`, `human`, `vehicle`, `low_confidence`, or a species name.
`confidence`	Classifier confidence score (0-1). `NA` for empty images.
`correct_label`	Correction entered during validation. Empty if the AI label was accepted.
`human_verified`	`TRUE` once a detection has been reviewed in the Validate tab.

Detection detail columns

Column	Description
`detector_label`	What the detector found: `animal`, `human`, or `vehicle`.
`detector_confidence`	Detector confidence score.
`cv_label`	

	Raw species label from the classifier before geofencing or thresholding.
`cv_confidence`	Classifier confidence for `cv_label`.
`cv_model`	Name of the classifier used, e.g. `deepfaune-v1.4`.
`bbox_left`, `bbox_top`, `bbox_right`, `bbox_bottom`	Bounding box coordinates. Normalised (0-1) when `bbox_normalised = 1`.

EXIF columns

wildtag automatically extracts camera metadata including `DateTimeOriginal`, `Make`, `Model`, GPS coordinates, `ExposureTime`, and `ISOSpeedRatings`.

> For analysis, use `correct_label` where populated, falling back to `label` where not. This gives the best available identification for every detection.

9. Privacy and GDPR

Camera traps occasionally capture people. wildtag handles this in two ways:

- **Detection labelling** - MegaDetector identifies humans and vehicles separately from animals, labelled `human` or `vehicle` in the CSV.
- **Automatic pixelation** - validation copies of human detections are heavily pixelated. Faces and bodies are unrecognisable. Original images are never modified.

> wildtag pixelates validation copies only. Original images remain unmodified and you remain responsible for their storage and access controls under applicable data protection law.

Appendix A: AI Models

MegaDetector v5a

Developed by Microsoft AI for Earth. A YOLOv5-based detector trained on millions of camera trap images worldwide. Identifies three categories: animal, human, and vehicle, with bounding boxes. Does not identify species.

- Architecture: YOLOv5x, 140M parameters, 1280px input
- Weight file: 280 MB

DeepFaune v1.4

Developed by CNRS (France). A vision transformer classifier trained on European camera trap images. Classifies animal crops into 38 classes covering the full range of European wildlife. Downloaded from the Models tab on first use (about 1.1 GB) and used offline afterwards.

- Architecture: ViT-Large with DINOv2 backbone, 182px crop
- Classes: 38 including domestic animals and birds as a group
- Weight file: 1.1 GB (downloaded on first use, not bundled)
- Best for: European surveys, ungulates and carnivores

SpeciesNet Global (Google, v4.0.3a)

Developed by Google. A complete end-to-end pipeline combining its own internal MegaDetector with a global species classifier and geofencing. Outputs taxonomy-aware predictions that roll up to family or order level when species-level confidence is insufficient.

- Architecture: EfficientNetV2-M classifier with internal MegaDetector
- Classifier weight file: 224 MB (downloaded on demand on first use, not bundled)
- Best for: global surveys or taxonomy-level rollup, **on a machine with a GPU** (very slow on CPU)

For UK and European surveys, DeepFaune v1.4 with the GBR geofence is faster and well-calibrated for local species. SpeciesNet is better for international datasets or when you want taxonomy-level predictions such as "cervidae family" rather than a forced species-level guess.

Appendix B: Geographic Filtering

The geographic filter reduces false positives by removing species predictions that are implausible for your study region. When GBR is selected, wildtag checks each prediction against a curated list of UK species. If the top prediction is not on the list, wildtag uses the next most confident plausible prediction instead.

UK species list includes

- **Deer:** red deer, roe deer, fallow deer, sika deer, Chinese water deer, muntjac
- **Carnivores:** red fox, badger, otter, stoat, weasel, polecat, pine marten, mink, wildcat
- **Rodents:** red squirrel, grey squirrel, water vole, bank vole, field vole, wood mouse, dormouse, brown rat
- **Lagomorphs:** rabbit, brown hare, mountain hare
- **Other:** hedgehog, mole, wild boar, all shrews, domestic livestock, dog, cat, bird

Broad group labels such as mustelid, lagomorph, and bird always pass the geofence since they are plausible anywhere. Select None to disable geofencing entirely.